STOCHASTIC DECISION MAKING

College Canteen Food Demand Optimization

A scenario-based stochastic optimization model for daily food preparation under uncertain demand

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Note: All numerical inputs are illustrative assumptions for demonstrating the model, not verified canteen transaction data.

Abstract and Introduction

Abstract. A college canteen must decide how many meals to prepare before the actual number of customers is known. Preparing too many meals can create leftovers and waste, while preparing too few can cause shortages and reduce customer satisfaction. This project formulates the decision as a stochastic optimization problem. It represents demand using discrete scenarios and probabilities, calculates profit under each scenario, and selects the preparation quantity that maximizes expected profit.

The prototype is implemented in Python using NumPy and Pandas. It evaluates all feasible integer preparation quantities up to the canteen capacity. Under the base assumptions, the model recommends 200 meals and estimates expected profit of ■5,325 per day.

1. Introduction

Food demand varies with attendance, class schedules, weather, campus events and customer preferences. The preparation decision must be made before demand is fully known. A fixed quantity may perform well on an average day but poorly when demand is unusually high or low. Stochastic decision making provides a way to evaluate multiple possible outcomes instead of relying on a single forecast.

2. Problem Statement

The canteen manager chooses a non-negative whole number of meals to prepare, subject to a maximum capacity. Actual customer demand is uncertain and is represented by possible demand levels and probabilities. The objective is to maximize expected daily profit after accounting for sales revenue, preparation cost, leftover recovery and shortage impact.

3. Objectives

- Represent demand uncertainty with scenarios and probabilities.
- Formulate a profit-maximization model with a decision variable and constraints.
- Implement an exhaustive search in Python.
- Compare stochastic and deterministic decisions.
- Study the impact of changes in costs and demand assumptions.

Scope, Assumptions and Dataset

4. Scope

The prototype models one food item for one operating day. Users can enter the food name, planned quantity, selling price, preparation cost, leftover value, shortage penalty, capacity, demand scenarios and probabilities. It calculates the profit for the user's chosen quantity and separately recommends the quantity with the highest expected profit.

5. Base Assumptions

Parameter	Base value	Description
Expected demand	200 meals	Weighted scenario average
Selling price	■70 / meal	Revenue per meal sold
Preparation cost	■40 / meal	Cost per meal prepared
Leftover recovery	■5 / meal	Value recovered per leftover
Shortage penalty	■10 / meal	Proxy for service impact
Capacity	300 meals	Maximum preparation quantity

6. Demand Scenarios

Scenario	Demand	Probability
1	150 meals	10%
2	180 meals	20%
3	200 meals	40%
4	220 meals	20%
5	250 meals	10%
Total	—	100%

Expected demand = 150(0.10) + 180(0.20) + 200(0.40) + 220(0.20) + 250(0.10) = 200 meals.

7. Synthetic Dataset

A 1,000-day synthetic dataset was generated by randomly sampling from the five demand outcomes using the stated probabilities and NumPy random seed 42. The sample mean was 199.14 meals. It differs slightly from the theoretical mean of 200 because a finite random sample varies. Synthetic data demonstrates the workflow but must not be presented as actual canteen records.

Mathematical Formulation

8. Decision Variable

Let Q be the number of meals prepared. Q is a non-negative integer and cannot exceed the canteen's maximum capacity.

9. Uncertain Demand

Let D■ denote demand in scenario s and p■ its probability. The scenario probabilities satisfy Σ p■ = 1. The base scenario set is {150, 180, 200, 220, 250}, with probabilities {0.10, 0.20, 0.40, 0.20, 0.10}.

10. Sales, Leftovers and Shortages

Meals sold = min(Q, D■)
Leftovers = max(Q – D■, 0)
Shortage = max(D■ – Q, 0)

11. Profit Function

Let r = ■70 be selling price, c = ■40 be preparation cost, v = ■5 be leftover recovery value, and h = ■10 be shortage penalty.

$$\pi(Q,D■) = r \cdot \min(Q,D■) + v \cdot \max(Q–D■,0) – cQ – h \cdot \max(D■–Q,0)$$

The first term is sales revenue, the second is leftover recovery, the third is the cost of all meals prepared, and the final term represents the modeled impact of unmet demand. The shortage penalty is a proxy for service or goodwill impact, not necessarily a direct cash expense.

12. Objective and Constraints

Maximize expected profit: $E[\pi(Q,D)] = \Sigma■\ p■ \times \pi(Q,D■)$.

Subject to: $0 \le Q \le 300$; Q is an integer; $p■ \ge 0$; and $\Sigma■\ p■ = 1$. The capacity is a base assumption and can be changed in the interactive application.

Optimization Method and Python Implementation

13. Method Selection

The project uses exhaustive search (enumeration). Since capacity is 300 meals, only 301 integer quantities are feasible. For every candidate Q, the program calculates profit under each demand scenario, multiplies each profit by its probability and sums the weighted profits. The quantity with the highest expected profit is selected.

This approach is transparent, easy to test and does not require a specialized optimization solver. Larger models with multiple food items, ingredients, days and shared resources may require a mathematical programming solver.

14. Implementation Steps

1. Import NumPy and Pandas.
2. Define demand scenarios and probabilities.
3. Generate a reproducible sample dataset.
4. Define price, cost, leftover value, shortage penalty and capacity.
5. Implement a profit calculation function.
6. Evaluate expected profit for every feasible Q.
7. Select the best quantity.
8. Run sensitivity analysis and deterministic comparison.

15. Core Python Logic

```
demand_values = [150, 180, 200, 220, 250]
probabilities = [0.10, 0.20, 0.40, 0.20, 0.10]

def calculate_profit(Q, D):
    sold = min(Q, D)
    leftover = max(Q - D, 0)
    shortage = max(D - Q, 0)
    return (sold*70 + leftover*5 - Q*40 - shortage*10)

results = []
for Q in range(301):
    expected = sum(p*calculate_profit(Q,D)
        for D,p in zip(demand_values, probabilities))
    results.append((Q, expected))

best_Q, best_profit = max(results, key=lambda x: x[1])
```

The complete notebook also contains data generation, exploratory analysis, test cases, sensitivity analysis and deterministic comparison.

Optimization Results

16. Base-Case Result

The exhaustive search evaluated every integer preparation quantity from 0 to 300. The best result occurred at Q = 200 meals. Expected profit was ■5,325 per day under the base assumptions.

Preparation quantity	Expected profit
190 meals	■5,035
200 meals	■5,325
210 meals	■5,215

The values illustrate the peak around 200 meals. The program evaluates all 301 feasible values, not only the three shown.

17. Scenario-Level Calculation at Q = 200

Demand	Probability	Profit	Weighted profit
150	10%	■2,750	■275
180	20%	■4,700	■940
200	40%	■6,000	■2,400
220	20%	■5,800	■1,160
250	10%	■5,500	■550
Total	100%	—	■5,325

For demand of 150 and preparation of 200, there are 50 leftovers. Recovery value is 50 × ■5 = ■250. Sales revenue is 150 × ■70 = ■10,500; preparation cost is 200 × ■40 = ■8,000; shortage is zero. Profit is ■10,500 + ■250 – ■8,000 = ■2,750.

18. Interpretation

The selected quantity balances the modeled cost of excess preparation against the modeled impact of unmet demand. It is not a claim that demand will always equal 200. It is the decision with the highest probability-weighted profit for the supplied scenario set.

Sensitivity Analysis

19. Purpose

Sensitivity analysis tests whether the recommendation changes when key assumptions change. It is important because the base prices, costs and scenario probabilities are estimates. The results indicate how dependent the decision is on those assumptions.

20. Results

Scenario	Changed assumption	Optimal Q	Expected profit
Base case	No change	200	■5,325.00
Higher prep cost	■40 → ■50	200	■3,325.00
Lower selling price	■70 → ■60	200	■3,415.00
Demand +10%	All demand values +10%	220	■5,857.50

21. Interpretation

Higher preparation cost: Expected profit falls by ■2,000, but the optimal quantity remains 200 meals. This demonstrates that a decision can remain stable even when profitability decreases.

Lower selling price: Expected profit falls to ■3,415 and the optimal quantity remains 200. Lower prices reduce the return from each sale.

Demand increases by 10%: The optimal quantity rises to 220 meals. Higher demand makes preparing more meals attractive.

22. Limitations of the Tests

Each test changes one assumption while holding the others fixed. Real-world costs and demand may change together. The result also depends on the selected scenarios and probabilities. Sensitivity analysis is therefore an illustration of model behavior, not a guarantee of future performance.

Deterministic Comparison and Web Application

23. Deterministic Model

The deterministic comparison assumes demand is known and fixed at exactly 200 meals. It evaluates each preparation quantity and selects Q = 200, with profit of ■6,000.

24. Comparison

Aspect	Deterministic	Stochastic
Demand assumption	Exactly 200	Five possible outcomes
Optimal quantity	200 meals	200 meals
Profit measure	Profit at demand 200	Expected profit across scenarios
Profit	■6,000	■5,325
Uncertainty	Not modeled	Explicitly modeled

Both methods recommend 200 meals in this base case. However, ■6,000 is conditional profit when demand is exactly 200, while ■5,325 is probability-weighted expected profit across all scenarios. These are different measures and should not be treated as identical.

25. Interactive Web Application

A browser-based prototype accompanies the project. It accepts the food name, planned quantity, capacity, selling price, preparation cost, leftover recovery value and shortage penalty. Users can add or remove demand scenarios and enter their own probabilities.

After validation, the application calculates expected profit for the user's quantity, evaluates all feasible quantities and displays the optimal recommendation, a profit-versus-quantity chart and scenario-wise results. It checks that demand values are whole numbers and that probabilities total 100%.

26. User Workflow

The canteen manager enters a menu item and financial details, adds possible demand outcomes with probabilities, and runs the calculation. The dashboard compares the manager's planned quantity with the model recommendation. The user can change assumptions and recalculate.

Evaluation, Limitations and Future Work

27. Model Evaluation

The profit function was checked using a leftover case and a shortage case. For Q = 100 and D = 80, profit is ■1,700. For Q = 100 and D = 120, profit is ■2,800. These test cases check that leftover recovery and shortage penalty are applied in the correct situations.

At Q = 200, the five weighted scenario profits sum to ■5,325. The deterministic case at demand 200 produces ■6,000. The enumeration checks every feasible integer quantity, so the selected value is optimal within the stated capacity and assumptions.

28. Limitations

- Demand scenarios and probabilities are illustrative, not estimated from actual point-of-sale records.
- The model considers one food item and one planning period.
- Per-meal costs and leftover recovery are assumed constant.
- Shortage penalty is a simplified proxy for customer dissatisfaction or lost goodwill.
- Ingredient inventory, staff availability, storage limits, expiry, supplier lead times and food safety are not explicitly optimized.
- Synthetic observations cannot replace historical operating data.

29. Future Enhancements

The next step is to collect actual daily sales and leftover data. Scenario probabilities can then be estimated from historical records, with separate demand patterns for weekdays, weekends, examinations and campus events. A multi-item model could allocate shared ingredients and kitchen capacity across menu items. Further extensions could include uncertain ingredient prices, perishable inventory, rolling daily re-optimization and a database-backed dashboard.

30. Responsible Use

The model should support, not replace, canteen staff judgement. Food safety and quality requirements must take priority over recovering value from leftovers. A real deployment should use validated costs, realistic demand estimates and safe handling or disposal policies.

Conclusion and References

31. Conclusion

This project demonstrates how stochastic optimization can support a college canteen's daily food preparation decision. It represents uncertain demand with discrete scenarios and probabilities, defines a profit function that accounts for sales, preparation cost, leftover recovery and shortages, and maximizes expected profit subject to preparation capacity.

Under the illustrative base assumptions, the model recommends 200 meals and calculates expected profit of ■5,325. Sensitivity analysis shows that a 10% increase in demand changes the recommendation to 220 meals. The deterministic comparison also recommends 200 meals, but its ■6,000 profit assumes demand is exactly 200 and is not the same as stochastic expected profit.

The interactive prototype allows users to enter their own food, cost and demand assumptions. The most important improvement for practical accuracy is to replace illustrative assumptions with real canteen data.

32. References

[1] S. M. Ross, *Introduction to Probability Models*, 12th ed., Academic Press, 2019.

[2] D. P. Bertsekas, *Dynamic Programming and Optimal Control*, 4th ed., Athena Scientific, 2017.

[3] NumPy Developers, “NumPy Documentation.” https://numpy.org/doc/ (accessed October 2026).

[4] pandas Development Team, “pandas Documentation.” https://pandas.pydata.org/docs/ (accessed October 2026).

[5] Python Software Foundation, “Python 3 Documentation.” https://docs.python.org/3/ (accessed October 2026).

33. Appendix: Reproducibility Details

Demand values: [150, 180, 200, 220, 250]. Probabilities: [0.10, 0.20, 0.40, 0.20, 0.10]. Financial parameters: selling price ■70, preparation cost ■40, leftover value ■5, shortage penalty ■10. Capacity: 300 meals. Synthetic data: 1,000 observations, NumPy random seed 42.

Associated project files: Python notebook, generated demand CSV, interactive HTML prototype and this report. Add exact charts exported from the completed notebook before final submission if required by the course instructor.

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